

Design and Conversion of Ink Brushstrokes in Three-dimensional Space

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Figure 1: Our 3D ink works titled 'Bird,' 'Gourd,' and 'Cat'

Abstract

Ink painting, as a highly esteemed form of traditional Eastern art, has been increasingly recognized for its 3D design form due to the growing demand for 3D digital materials in the gaming and

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animation industries. However, creating artistically appealing 3D ink paintings is a considerable challenge, especially in replicating the delicate effects of two-dimensional brush strokes in three-dimensional space. Most current research mainly focuses on using rendering techniques to simulate the effect of ink, with a relative lack of exploration in design methods for 3D transformation of ink paintings. In this paper, we present three 3D ink works created by us, emphasizing the study of the core element of ink painting: the "brush stroke," and its restoration and expression in three-dimensional space. We have outlined design dimensions based on brush stroke characteristics and employed a combination of brushes and procedural shader to recreate the features of brush strokes in three-dimensional space.

CCS Concepts

• **Computing methodologies** → **Non-photorealistic rendering**.

Keywords

Chinese painting, 3D Art, Non-photorealistic rendering

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1 Introduction

Ink painting is an important art form that uses black ink and white space to convey the essence of objects. It has a unique aesthetic that presents abstraction and flatness through brush strokes and ink tones. The ink painting style has become a distinctive art form worldwide, and with the continuous evolution of media, ink painting animations and games are gradually emerging. By leveraging digital technology, the ink painting aesthetic can be recreated in various forms, achieving cultural innovation and offering a unique visual experience.

In recent years, the emergence of 3D media such as VR and gaming has presented significant opportunities and demands for 3D ink painting design [Guo and Chung 2019; Wang et al. 2018]. However, there is a conflict between the abstraction and simplicity of ink painting and the concrete physicality of three-dimensional space. The challenge is to ensure that ink paintings can be viewed from any angle in 3D space while maintaining their original beauty. Transforming ink paintings into 3D is a highly challenging endeavor with many difficulties and obstacles. Although many studies and works have attempted this transformation, few have approached it from a design perspective.

We conducted a hierarchical study of these challenges and, through the analysis of classic works and a review of the literature, proposed a method for transforming brush strokes based on three-dimensional visual features. We re-examined issues such as the composition, applicability, and ambiguity of ink and brush strokes, and concluded that the core issue is that three-dimensional objects depicted with ink brush strokes require different functional brush stroke designs, which must work together as a whole to achieve a combined advantage. Based on this, we focused on the design of brush strokes for the surface, edges, and space of three-dimensional objects. We developed corresponding creation tools in Blender and completed three works (Figure 1). This paper elaborates on the aforementioned methods by incorporating specific details from the creation process of the work, and discusses and reflects from the perspective of a creator and developer.

2 Related Works and Key Issues

Research on ink painting brush strokes has a long history, with diverse research paths existing alongside each other. Particularly in recent years, innovative research on ink painting brush strokes has yielded many exciting results. In the field of computer graphics, a number of attempts using stylized rendering algorithms have emerged [Deng Siyu 2017; Sun et al. 2010; Yuan et al. 2007]. These studies can automatically render 3D models into ink painting effects

from various angles, but they often suffer from long rendering times and lack of aesthetic appeal and brush stroke detail. In artistic practice, many artists combine their modeling experience to create 3D ink paintings through layered 2D images or texture painting [Neoh et al. 2003; Zhang and Sun 2020]. While preserving the artist's essence, this approach is limited to specific viewing angles and struggles to present the three-dimensional texture of brush strokes. Additionally, some VR artists have used tools like Tilt Brush to create many ink painting-style works [Google 2022], spending considerable effort to simulate the texture of ink brush strokes through repeated painting—black-and-white paintings with brush marks. However, this often overlooks the inherent characteristics of ink painting, such as the specific rules and patterns of brush strokes.

After reviewing related research and works, we summarized the following challenges:

- Firstly, the brush strokes in traditional ink painting are a planar abstraction of depicted objects. Consequently, three-dimensional spaces and objects based on perspective are not directly suitable for these brush strokes. When flat brush strokes enter three-dimensional space, they often become distorted, failing to meet the multi-angle viewing requirements of 3D space and losing their shaping capability.
- Secondly, traditional ink painting covers a wide range of subjects, including landscapes, flowers, birds, and figures. Some brush strokes are specifically suited for certain subjects; for example, fluffy brush strokes are often used to depict the texture of bird feathers. These brush strokes demand high fidelity in texture and detail reproduction. When flat brush strokes enter three-dimensional space, significant detail is often lost, requiring reasonable imagination and design to fill the gaps and ensure the textural integrity of the 3D brush strokes.
- Lastly, traditional ink painting is highly spiritual and arbitrary, not pursuing precise depiction of objects. For instance, the edges and contours of objects are blurred and unclosed. This trait contradicts the clarity required in three-dimensional space. In two-dimensional ink paintings, the contours of objects are formed by the diffusion of water and ink on paper, and replicating this material characteristic in virtual space is highly challenging.

3 Brush Stroke Design Method

We believe that the visual style of traditional ink painting arises from the interplay of various factors, such as materials, themes, composition, and colors. However, the most prominent feature is the fusion of water and ink, creating strong contrasts and distinctive brush marks. Starting from visual outcomes, we analyzed numerous existing ink paintings and found that their brush strokes can be functionally divided into shaping brush strokes, outlining brush strokes, and coloring brush strokes (Figure 2). Shaping brush strokes can directly represent the drawing objects and are primarily used to depict flat objects such as petals, feathers, and beards. Outlining brush strokes and coloring brush strokes are often used together, with outlining brush strokes mainly used to delineate the edges of objects, generally exhibiting an intermittent effect following the shape's contours, while coloring brush strokes are used to fill in the

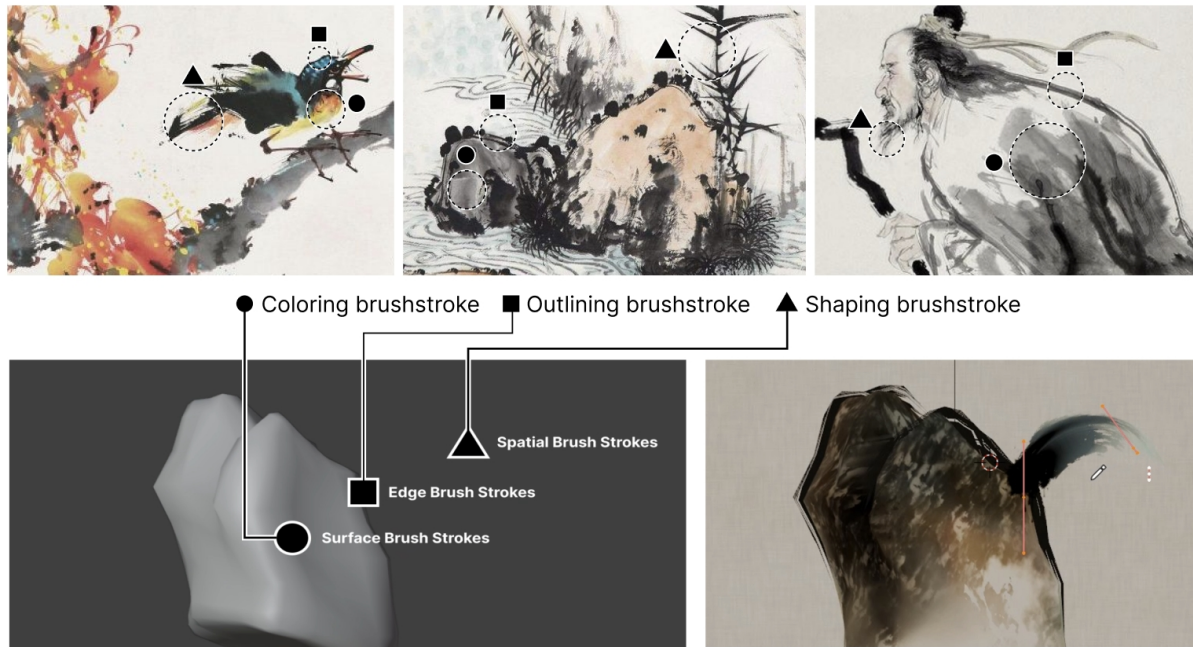


Figure 2: The corresponding relationships of three types of brush strokes in traditional Chinese painting (above) mapped onto three-dimensional models (below).

areas outlined by the outlining brush strokes, such as the texture of clothes or the folds of rocks.

In three-dimensional space, we designed and implemented spatial brush strokes, edge brush strokes, and surface brush strokes based on their relationship with the surface of the 3D models (Figure 2). In this chapter, we will demonstrate and explain each type of brush stroke in conjunction with our works.

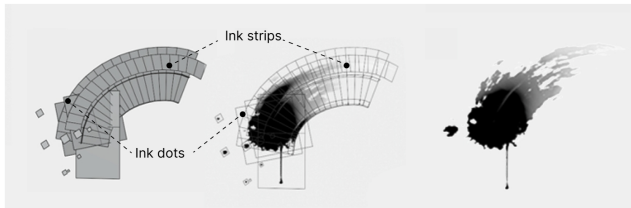


Figure 3: Spatial Brushstrokes: The Model (left), Wireframe (middle), and Rendering from different angle (right)

3.1 Spatial Brush Strokes

The core of the spatial brush stroke research lies in how to construct individual brush strokes in three-dimensional space that appear aesthetically pleasing from all directions. One stroke is primarily composed of ink dots at the head and ink strips at the tail as shown in Figure 3. We designed the ink strips as multiple intersecting flake-like models and used the billboard method to handle the ink dots at the head so that they always face the camera. These two methods

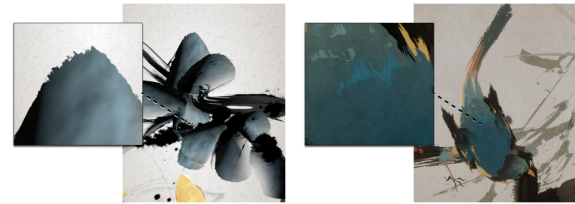


Figure 4: High moisture level details of 'The Gourd' and 'Bird'

complement each other effectively, ensuring that the stroke appears flat from all directions.

We've identified five key dimensions of ink technique: moisture, darkness, dryness, trailing, and color. These independent dimensions can be freely adjusted to create various ink-wash effects.

- **Moisture Level:** This simulates the brush's ability to absorb water. Lower values produce a rougher edge texture, whereas higher values generate a watercolor diffusion effect on the paper. In the artwork "Gourd," the leaves are rendered with a high Moisture Level, capturing their delicate freshness and vibrancy (Figure 4 left). Similarly, "Bird" features certain feathers rendered with this technique, primarily to express the soft, fluffy texture of the fine feathers (Figure 4 right).
- **Dryness:** Affects the hollowness at the center of the stroke, a common technique in ink painting. This characteristic is widely present in our various works, where different coefficients can express entirely different effects as listed in

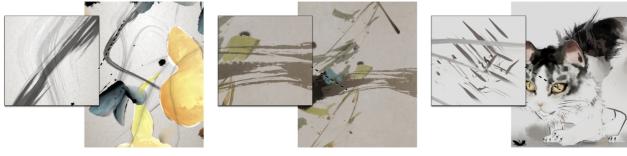


Figure 5: The branches of 'Gourd', the trunk of 'Bird', and the short fur of 'Cat'; from left to right, the dryness increases progressively

Figure 5. Higher dryness levels result in more pronounced hollowness, while lower levels create fuller strokes.

- **Trailing:** Controls ink distribution along the stroke. Higher values place more ink at the start, lower values concentrate it in the middle. "We extensively utilized varying trailing in designing the branches of 'The Gourd' to express different branch shapes and a sense of flexibility (Figure 6 left)."
- **Darkness:** Determines ink concentration. Higher darkness makes strokes blacker, lower darkness allows colors to stand out.
- **Color:** Incorporates up to four colors per stroke for gradient effects. "It is especially abundant and finely executed in the feathers of 'Bird,' employed to bring out the nuanced layers of the feathers (Figure 6 right)."

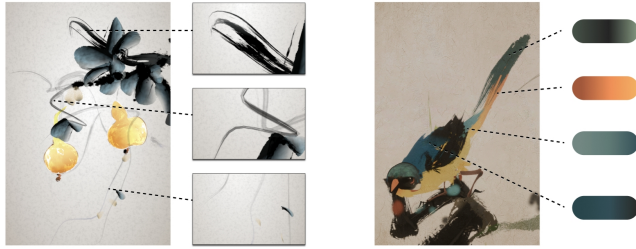


Figure 6: Different trailing in 'The Gourd' (left) and different color in 'Bird' (right)

In terms of geometric characteristics, we primarily design based on four aspects: size, length, thickness, and twist (Figure 7), enabling not only holistic adjustments to the brushstrokes but also modifications in different regions of the stroke for more intricate effects. Notably, thickness and twist have introduced a new dimension to artistic expression in 3D ink painting, a feat previously unattainable with traditional 2D media. We commonly use brushstrokes with zero thickness for depicting leaves or petals, and strokes with a thickness of 1 for branches. Twist is employed to depict completely twined vines.

3.2 Edge Brush Strokes

In traditional ink painting, objects with volumetric shapes are often outlined, as shown in Figure 8. In the cases of "Cat" and "Gourd," we first modeled the body of Cat and the fruit of the gourd, then expanded the model's normals outward to simulate the effect of outlining. To make the outlining effect more vivid, we applied a



Figure 7: The original brushstroke and the effects of adjusting the four dimensions separately.

separate material to this expanded model, controlled by a procedurally generated noise map to simulate the intermittent effect of brushstrokes. On the other hand, this method allows us to precisely control the thickness of the strokes (i.e., the extent of outward expansion), as seen in the smooth variations in thickness in "Cat" and "Gourd". It's worth mentioning that, to better integrate with the internal volume of the objects, we also overlaid a Fresnel effect on the original model, keeping the color of the outermost layer of Fresnel consistent with the color of the expanded normals, achieving a well-blended effect.

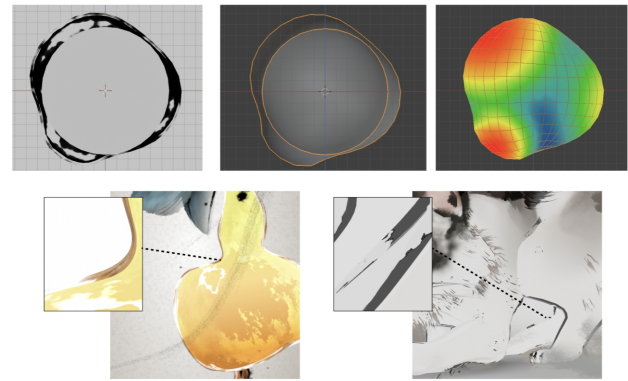


Figure 8: The principle of outlining (above) and the outlining in our work (below)

Our outlining technique for closed surfaces (Figure 8) involves creating an expanded outer shell of the original model with flipped normals, visible only from the back. We control outline thickness using vertex weights, allowing for varying thickness based on viewing angle. For shading, we focus on creating brushstrokes that follow the edge flow. We obtain gradient information by multiplying the model's normal direction with the camera direction, then use this to sample a noise texture, creating stripes parallel to the model's edges. By overlaying another noise texture sampled using world space coordinates, we achieve a vivid brushstroke effect with intermittent continuity and disjointedness, effectively mimicking traditional ink painting techniques in 3D space.

3.3 Surface Brush Strokes

The method for surface brush strokes is similar to traditional texturing, where a two-dimensional image is attached to the surface of the model. As shown in the Figure 9, we demonstrate the complete creation process of a surface brush stroke. Our tool automatically

creates a background texture to simulate rice paper. Users can select the area to be covered by the brush stroke texture through drawing and adjust the size and intensity of the brush stroke using parameters. Additionally, the base image with brush strokes can be blended with gradient colors or the original color map to create colorful results. In the figure, the gourd is made using a large brush stroke and gradient overlay method, the bird's feathers are created with small brush strokes and gradients, while the cat's face blends a drawn ink pattern.

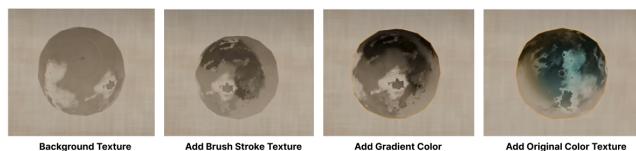


Figure 9: The complete creation process of brush stroke on surface.

Surface brush strokes are fully procedural, generated based on different combinations of noise types, ensuring a random and lively appearance. For the brush strokes, we first use vertex colors to control the areas where the brush strokes appear, allowing users to paint on any model without needing UV mapping. We generated various types of seamless basic brush stroke textures using AI, which are then mapped onto the model using a multi-view projection method and displayed in combination with vertex colors. When blending brush stroke textures with color maps, we use a blur and hue mixing mode to achieve the ink diffusion effect.

4 Discussion

How to present the beauty of ink painting in three-dimensional space is a question that our research continuously focuses on. It involves the integration and transformation of various aspects of art, technology, and design, lying at the intersection of traditional art and modern forms, artistic expression and technical adaptation, and the ontology of art itself. Our research has explored and practiced the transformation of the core element of ink painting—brush strokes—into three-dimensional space at multiple levels. We aim to provide a solution that differs from existing research approaches and technical solutions, intending to help artists, designers, or any creators interested in ink painting to freely express their artistic creativity in three-dimensional space, thus extending the beauty of ink painting to a broader, new field.

During the research process, we found that the 3D transformation of ink painting brush strokes represents a promising area of study. More broadly, it addresses the question of whether aesthetics can transcend boundaries. Classical art, with its rich heritage, has formed highly autonomous aesthetic standards and awareness. Whether this aesthetic experience and standard can transcend time and media to enter the digital world, and even bridge the gap between two-dimensional and three-dimensional spaces, is a challenge faced by many art forms. Notably, the authors are both developers of 3D brush strokes and accomplished creators. Benefiting from the accumulation of experience in multiple roles, we discovered several thought-provoking phenomena:

The Whole and the Part: We strive to achieve a "natural feel" in ink paintings drawn with 3D brush strokes, where the whole and the details are harmonious and integrated, imbued with liveliness and vividness. For example, we can view the feathers covering a bird as a whole, composed of individual feathers, each rendered with different brush strokes to create a rich and varied visual effect. To this end, we use procedural and randomization techniques to enhance local diversity, while following the principle of regularity to ensure the integrity of the whole. Particularly when presenting textured materials, it is crucial to coordinate global and local coordinates. As the renowned painter William Hogarth advocated in "The Analysis of Beauty," we strive to apply the principle of "variety in unity" in 3D space to preserve the beauty of flat ink paintings.

Blending and Cutting: When flat paintings transition to three-dimensional forms, they inevitably face the issue of "boundaries." This is especially true when traditional ink paintings, which do not adhere to geometric spatial views, enter three-dimensional space, potentially distorting or losing original boundaries and contours while creating new boundaries and overlaps. We found that when mapping the textures of traditional ink paintings into 3D space, surface brush strokes and spatial brush strokes often need to be blended to highlight a sense of integration. Conversely, edge brush strokes need to be cut to connect surface brush strokes with spatial brush strokes. Accumulating such creative experiences may provide valuable support for 3D ink painting to become a new art form.

Reality and Reproduction: For a long time, the accurate representation of reality has been a significant topic in the art world, and this issue also appears in the 3D transformation of ink painting brush strokes. For instance, when developing brush strokes commonly used for "Bird", we referenced many real bird images and tried to decode the bird the artist saw in ancient paintings. With this understanding, we then reproduced the bird in ancient paintings using 3D ink brush strokes. This transformation indeed helped us gain a deeper understanding of the artist's thoughts and state while painting, which will significantly impact the future development and innovation of brush strokes.

5 conclusion

Our research focuses on the significant challenges faced in creating visually captivating 3D ink paintings. Approaching from an innovative design perspective, we first delve deeply into the understanding and importance of brush strokes. Subsequently, we elucidate our design methodology through specific design works. This study offers an approach to aesthetic transcendence, reconstructing the two-dimensional artistic language into three dimensions. Based on the creative results of numerous users, this attempt not only preserves the aesthetic appeal of ink painting but also reveals new three-dimensional visual interests. We believe that more attempts and explorations should be conducted in the future, as there are still many artistic forms and aesthetic experiences awaiting discovery.

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